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An Overview of Vehicular Platoon Control under the Four-Component Framework

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Abstract— **The platooning of autonomous ground vehicles has potential to largely benefit the road traffic, including enhancing highway safety, improving traffic utility and reducing fuel consumption. The main goal of platoon control is to ensure all the vehicles in the same group to move at consensual speed while maintaining desired spaces between adjacent vehicles. This paper presents an overview of vehicular platoon control techniques from networked control perspective, which naturally decomposes a platoon into four interrelated components, *i.e.*, 1) node dynamics (ND), 2) information flow topology (IFT), 3) distributed controller (DC) and, 4) geometry formation (GF). Under the four-component framework, existing literature are categorized and analyzed according to their technical features. Three main performance metrics, *i.e.* string stability, stability margin and coherence behavior, are also discussed.**

I. INTRODUCTION

The platooning of autonomous vehicles on highway have attracted extensive interests due to its potential to largely benefit road traffic, *e.g.* enhancing highway safety, improving traffic utility and reducing fuel consumption [1]. The control of autonomous platoon aims to ensure all the vehicles in the same group to move at consensual speed while maintaining the desired spaces between adjacent vehicles [4].

The earliest implementation can date back to the PATH program during the last eighties in California, in which many fundamental topics were studied, including major goal of platooning, division of control tasks, layout of control architecture, technologies for sensing, actuation and communication, as well as control laws for headway control, *etc.* [2]-[6]. Since then, many issues on platoon control have been discussed and addressed, such as the selection of spacing policies [7]-[9], how to consider powertrain dynamics [10], homogeneity and heterogeneity [11]-[14] *etc.* Advanced control methods were also introduced into platoon control to achieve better performances. For instance, Barooah *et al.* (2009) introduced a mistuning-based control method to improve the stability margin of vehicular platoon [16]. Dunbar and Derek (2012) proposed a distributed receding horizon controller and derived the sufficient conditions to ensure string stability [18]. Ploeg *et.al.* (2014) developed a $\mathcal{H}_\infty$ control method, in which the string stability was explicitly satisfied [20]. More recently, some demos of platoon control have been performed in the real world, including the GCDC in the Netherlands [19], SARTRE in Europe [21], and Energy-ITS in Japan [22], *etc.*

The earlier platoon often only relies on radar-based sensing systems, in which the type of information exchange topologies is quite limited [23]. The rapid deployment of vehicle-to-vehicle (V2V) communications, such as DSRC and VANET [24], however, can generate a more variety of topologies for platoons, *e.g.* two-predecessor following type and multiple-predecessor following type [23][25]. New challenges naturally arise due to the variety of topologies, in particular when considering the time delay, packet loss, and quantization error in the communications. In such cases, it is more preferable to view the vehicular platoon as a network of dynamical systems, and to employ networked control perspective to design the distributed controllers. For example, Oncu *et al.* (2014) investigated the effect of network-induced constant delay and sampling-hold process on string stability from the networked control perspective [27]. Bernardo *et al.* (2014) proposed a method to analyze the platooning problem from the viewpoint of consensus control of a dynamic network, and proved the stability of platoon in the presence of time delay [29]. Wang *et al.* (2014) introduced a weighted and constrained consensus seeking framework to study the influence of time-varying network structures on the platoon dynamics by using a discrete-time Markov chain [28].

From networked control perspective, a vehicular platoon is actually one dimensional network of dynamical systems, in which the vehicles only use their neighborhood information for controller design but need to achieve the global coordination. The perspective naturally decomposes a platoon system into four interrelated components, *i.e.*, node dynamics (ND), information flow topology (IFT), distributed controller (DC), and formation geometry (FG) [28]-[30]. This decomposition is able to provide a unified four-component framework to analyze, design and synthesize vehicular platoon, as well as further on-road implementation [31]-[34]. Under the four-component framework, this paper summarizes the existing platoon control outcomes by literature categorization and technical analyses. Moreover, this paper also reviews the techniques dealing with three major performance metrics, *i.e.* string stability, stability margin and coherence behavior. The remainder of this paper is as follows: Section II introduces the four-component framework of a platoon system; Section III reviews the mainstream

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techniques by each component, followed by a brief of performance metrics in Section IV. Section V concludes this paper.

II. Introduction to four-component framework

This paper considers a platoon on a flat road (see Fig. 1), which aims to move at consensual speed while maintaining desired space among vehicles. The platoon has a leading vehicle (LV, indexed by 0) and other following vehicles (FVs, indexed from 1 to N). As demonstrated in Fig. 1, the platoon system can be viewed as a combination of four main components: 1) node dynamics; 2) information flow topology; 3) distributed controller, and 4) formation geometry. The four-component framework of a platoon is demonstrated in Fig. 1, which includes:

1) Node dynamics (ND), which describes the behavior of each involved vehicle;
2) Information flow topology (IFT), which defines how the nodes exchange information with each other;
3) Distributed controller (DC), which implements the feedback control only using neighboring information;
4) Formation geometry (FG), which dictates the desired inter-vehicle distance when platooning.

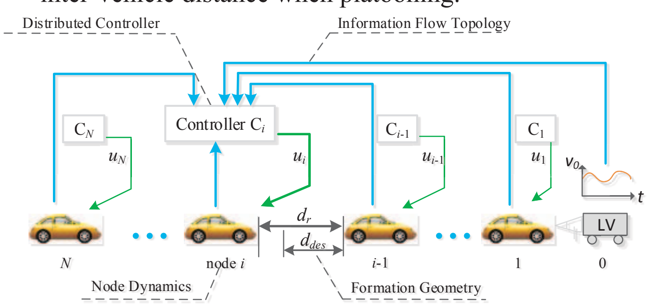

Fig. 1 Four major components of a platoon : 1) node dynamics, 2) information flow topology, 3) distributed controller, 4) geometry formation; where d_r is the actual relative distance, d_{des} is the desired distance, u_i is the the contol signal for i-th vehicle, and C denotes the controller.

The internal stability should be ensured for all platoons. A platoon is said to be internally stable if and only if the closed-loop system has eigenvalues with strictly negative real parts [32]. Besides internal stability, the most concerned performance metrics for a platoon are 1) string stability; 2) stability margin, and 3) coherence behavior:

Definition* 1 *(String Stability). A platoon is said to be string stable if and only if the disturbances are not amplified when propagating downstream along the vehicle string [11][35][41];

Definition* 2 *(Stability Margin). The stability margin of a platoon is defined as the absolute value of the real part of the least stable eigenvalue, which characterizes the convergence speed of initial errors [32][33];

Definition* 3 *(Coherence Behavior). The coherence behavior is quantified as the $\mathcal{H}_2$ norm of the closed-loop system [51], which is a scalar index that captures the robustness of a platoon subject to exogenous disturbances.

Each component in Fig. 1 has significant influence on the performance metrics of a platoon. The categorization result of existing literature is shown in TABLE I.

III. Literature Review on Each Component

A. Node Dynamics (ND)

Most research on platoon control only gives emphasis on the dynamical behaviors of ND in longitudinal direction. Only a few studies discuss the integrated longitudinal and lateral control [56][57]. The later control usually adopted the bicycle model, see [56][57] for details. This paper only reviews the modeling of ND in longitudinal direction.

The vehicle longitudinal dynamics are inherently nonlinear, which is composed of engine, drive line, brake system, aerodynamics drag, tire friction, rolling resistance, gravitational force, *etc.* [4][18]. Some studies directly use nonlinear models for platoon control, see [18][35][43] and [56] for examples,. The asymptotic stability and string stability can be guaranteed by carefully selecting the control parameters, but explicit performance limits are rather difficult to analyze with given spacing policy and communication topology. Actually, linear models are more frequently used for tractable issues. The commonly used models include 1) single integrator model, 2) second-order model (including double-integrator model), 3) third-order model, and 4) single-input-single-out (SISO) model (see TABLE I. for details).

The single integrator model is the simplest case, which takes the vehicle speed as control input and position as the exclusive state. This can significantly simplify the theoretical analysis on controller design. For instance, the structured optimal control of platoon can be transformed into a convex optimization problem under single integrator assumption, but quite challenging for other models[49][51]. However, besides largely depart from actually vehicle dynamics, the single integrator model fails to reproduce the slinky-type effects or string instability [44]. An improvement is to assume ND as a point mass, resulting in the double-integrator model [15][16][44], or even to consider a platoon as a mass-spring-damper system, resulting in the second-order linear model [69][72][73]. The two models both use acceleration as control input. Many important theoretical results, like decentralized optimal control [49], stability margin scaling trend [16][32][33][47][48], and coherence behavior [66], rely on the assumption of second-order dynamics. This assumption still does not catch many features of vehicle dynamics, *e.g.* inertial delay in powertrain dynamics, and might lead to instability in real world [10][17][19][20]. One modeling trend for ND is to further increase one state and yield so-called third-order model. The increased state is often used to approximate the input/output behaviors of powertrain, which equivalently degrades the control input to engine torque and/or braking torque [8][10][14][17][20]. Now the majority of approximation use either feedback linearization technique [3][10][17][23] or lower-layer control technique [2][4][56]. The last, but not the least, model is the SISO model, which is often used to analyze string stability from frequency domain. The pioneer work on this model started from Seiler, Pant, and Hedrick [41], and later widely employed in many other studies, *e.g.* [12][59][64] and [65].

TABLE I. CATEGORIZATION OF PLATOON CONTROL

Node Dynamics (ND)				
Single-integrator	Second-order model	Third-order model	SISO model	Nonlinear model
[49][50][51][52][53]	[13][15][16][28][29][32][33][44][46][47][48][49][61][62][63][66][69][72][73]	[3][5][6][8][9][10][11][14][15][17][19][20][23][26][27][30][31][34][36][42][45][58][67][68][70][71]	[12][41][59][64][65]	[18][35][43][54][56][60]
Homogeneous	Heterogeneous			
[3][5][6][8][10][15][16][20][23][28][30][31][32][33][34][41][42][46][48][49][50][51][52][53][58][59][61][62][63][65][66][69][70][73]	[9][10][11][12][13][14][17][18][19][26][27][29][35][36][43][44][45][47][54][56][60][64][67][68] [71][72]			

Information Flow Topology (IFT)				
PF	PFL	BD	BDL	TPF
[3][8][9][10][11][15][19][20][23][27][30][36][41][42][45][46][48][54][58][61][65][70][71][73]	[5][6][11][14][17][18][23][26][30][35][41][56][58][67][69]	[12][13][16][23][30][32][33][34][41][43][47][48][49][59][60][64][68][72][73]	[12][23][30][34]	[20][23][30][58]
TPFL	Undirected	Limited Range	General Topologies	
[23][30]	[23][30][31][34][44][62]	[64][66]	[23][28][29][30][32][50][51][52][53][63]	

Distributed Controller (DC)				
Linear Controller	Optimal Controller	$\mathcal{H}_\infty$ controller	SMC	MPC
[3][6][9][11][12][13][16][19][23][27][28][29][30][31][32][33][34][36][41][42][44][46][47][48][50][52][53][58][59][60][61][62][64][65][66][68][69][70][71][72][73]	[15][17][49][51][54][63]	[14][26][20][67]	[5][8][10][35][43][56]	[18][19][45]

Formation Geometry (FG)		
CD	CTH	NLD
[3][5][6][9][11][12][13][14][16][17][18][23][26][28][30][31][32][33][34][35][41][43][44][46][47][48][49][50][51][52][53][55][56][58][59][61][62][63][64][66][67][68][69][72][73]	[9][10][15][19][20][27][29][36][42][45][54][60][64][65][71][73]	[8][70]

Performance Metric		
String Stability	Stability Margin	Coherence Behavior
[3][5][6][8][9][10][11][12][14][15][17][18][19][20][26][27][35][36][41][42][43][44][45][46][54][56][58][59][60][61][62][64][65][67][68][69][70][71][72][73]	[13][16][23][28][29][30][31][32][33][34][47][48][63]	[16][49][50][51][52][53][66]

An important terminology for ND is homogeneity. A platoon is said to be "homogeneous" if all nodes have identical dynamics; otherwise, it is called to be "heterogeneous". The homogeneous assumption can simplify the theoretical analysis of platoon control (see [20][23][32] and [41] for examples), while the heterogeneous assumption is more aligned with the reality (see [9]-[12] and [17] for examples).

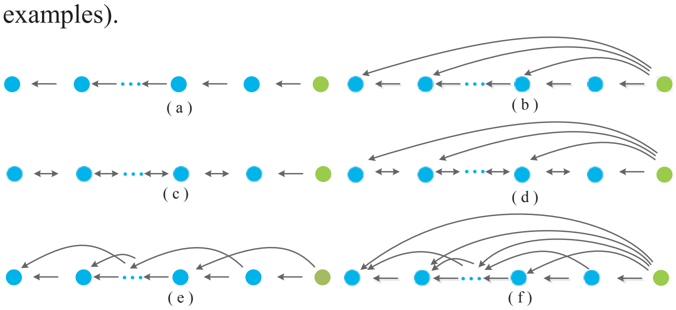

Fig. 2 Typical IFTs for Platoon. (a) PF; (b) PLF; (c) BD; (d) BDL; (e) TPF; (f) TPLF

B. *Information Flow Topology (IFT)*

The IFT applied in a platoon is closely related to the way a vehicle acquires the information of its surrounding vehicles. The IFT describes the information used by each local controller and has significant influence on the collective behaviors of the platoon, *e.g.* string stability [41], stability margin [30][32] and coherence behavior [49][56] *etc.*

Early-stage platoons are mainly radar-based, which means that a vehicle can only obtain the information of its nearest neighbors, *i.e.* front and back. Under this sensing system, IFTs are the predecessor following (PF) and bidirectional (BD) topologies (see TABLE I. for details). Nowadays, as the rapid deployment of vehicle-to-vehicle (V2V) communications, such as DSRC and VANET, various IFTs are emerging, including predecessor-following leader (PFL) type, two predecessor-following (TPF) type, two predecessor-following leader (TPFL) type, undirected, and other limited communication range topologies (see TABLE I. for details). Fig. 2 demonstrates some of these topologies. Note that TABLE I. is only based on the connection characteristics without consideration of the communication characteristics such as quantization, data dropout and time delay.

No matter what kind of topology is employed, internal stability must be guaranteed in a platoon. Two main approaches have been proposed to ensure the internal stability: 1) global approach, *e.g.* [14] and [67], and 2) local approach, *e.g.* [8][10][20] and [27]. The first approach is to straightforwardly take the overall platoon as a structured system and then design a centralized controller, for which IFT becomes less important to controller design. For example, the linear matrix inequality (LMI) was obtained based on the global platoon dynamics to guarantee internal stability in [14] and [67]. One major drawback for this approach is that the computation efficiency quickly worsens with increasing platoon size. Therefore, most studies decomposed a platoon into sub-systems and tried to use decentralized control methods, leading to the second approach. For instance, under PF topology, a platoon can be naturally viewed as unidirectional cascade systems, which only needs to study any two successive vehicles to guarantee stability, *e.g.* [8][10][20][27] and [36]. Besides, the inclusion principle was used to decompose such kind of platoon into locally decoupled subsystems, for which overlapping controller could be designed [17]. This decomposing technique does not suit for a platoon with BD topology because its spacing errors propagate from both forward and backward directions. The partial differential equation (PDE) technique were employed to approximate the dynamics of platoons under BD topology in [13][16][32][33] and [68], which avoided to analyze high dimensional dynamics. For platoons under general topologies, the matrix similarity transformation and factorization is an important approach, which actually decompose the internal stability into two components: 1) stability of information flow for given IFT and 2) stability of individual vehicles for given DC, see [23][30] and [55]. This method is only applicable for homogeneous platoon.

C. Distributed Controller (DC)

The DC implements the feedback control using neighbor's information to achieve the global coordination of the platoon. An unstructured DC is one that corresponds to complete graph which requires communication between any pair of vehicles. Many existing studies belong to structured control law either in an explicit or implicit way, see [9][10][17] and [49]. It is the structural property governed by IFT that brings both the difficulties in controller design [49] and fundamental limits in platoon performances [30][32][56].

The commonly used DC are linear for the purpose of comprehensive results on theoretical analysis, and convenience in hardware implementation [7][11][16][23]. The internal stability of linear controller largely depends on the structure of IFT, which means linear DC design is often in a case-by-case way. The stabilized region of linear control gains is explicitly derived in [23] for a large amount of topologies, and the string stability requirements for platoon under PF topology are established in [10]. The optimization methods, either numerical or analytical, were also proposed to optimize the localized gains in [15][17] and [49]. There are also some studies employing sliding model control (SMC) to design string-stable platoon [5][8][10]. For SMC, the internal stability and string stability of platoons have to be realized through a posterior controller tuning.

There are two main drawbacks in the aforementioned design methods, *i.e.* 1) unable to explicitly handle the string stability, and 2) unable to handle the state or control constraints. Recently, the $\mathcal{H}_\infty$ controller synthesis is proposed to include the string stability requirement as a priori in the design specification [20]. In addition, model predictive control (MPC) has been introduced into the platoon control to forecast system dynamics, explicitly handling actuator/state constraints by optimizing given objectives [18][19][45].

D. Formation Geometry (FG)

There are three major policies of FG for vehicular platoons: 1) constant distance (CD) policy, 2) constant time headway (CTH) policy, and 3) nonlinear distance (NLD) policy [7][58]. For the CD policy, the desired distance between two consecutive vehicles is independent of vehicle velocity, which can lead to a very high traffic capacity. For the CTH policy, the desired inter-vehicle range varies with vehicle velocity, which accords with driver behaviors to some extent but limits achievable traffic capacity. For the NLD policy, the desired inter-vehicle is a nonlinear function of vehicle velocity, which has the potential to improve both the traffic flow stability and traffic capacity compared with CD and CTH policies (see [8][70] for details).

IV. LITERATURE REVIEW ON PLATOON PERFORMANCE

Some practical benefits brought by platooning, such as reducing fuel consumption and improving traffic efficiency, are not covered in this paper. For these topics, interested reader can refer to [37]-[40]. The major techniques for internal stability are presented in Section II.A. Hence, this section only focuses on string stability, stability margin and coherence behaviors.

A. String Stability

Internal stability of a platoon in the Lyapunov sense does not guarantee string stability. If not well designed, error signals can amplify when propagating downstream the vehicle string, which may result in rear-end collision [10][41]. This effect is called string instability, *e.g.* in [41] or slinky effect, *e.g.* in [17].

The achievability of string stability has tight relationship with FG and IFT employed in the platoon. Seiler *et al.* (2004) showed that due to the complementary sensitivity integral constraint, string stability cannot be guaranteed for any linear identical controllers under PF topology and CD policy [41]. Barooah *et al.* (2005) further pointed out that for a homogeneous platoon under BD topology, linear identical controllers also suffered fundamental limitations on the string stability due to amplified spacing errors and disturbances [59]. Middleton *et al.* (2010) extended the work in [41] by considering heterogeneous ND, limited communication range, non-zero time headway policy, and showed that both forward communication range and small time headway cannot alter the string instability [64]. Some solutions are proposed to improve string stability, including:

a) Relaxing formation rigidity, *i.e.* introducing enough time headway in the spacing policy (*e.g.* [9][10][15]), or using nonlinear policy (*e.g.* [8][70]);
b) Using non-identical controller for different vehicle, (*e.g.* [16][46]);
c) Extending the information flow by using more complex IFT, *e.g.* broadcasting the leader's information (*e.g.* [11][17]). Note that that the analysis in [63] pointed out the necessity to have some global information (*e.g.* leader's velocity) in DC to ensure string stability.

Recently, some advanced controllers have been proposed to ensure string stability, including SMC [8][10], MPC [18][19] and $\mathcal{H}_\infty$ controller [14][20][67]. Note that all of them either employ CTH policy or use certain global information.

B. Stability Margin

Stability margin is used to characterize the convergence speed in a platoon [32][33]. Most of current research on stability margin focus on the CD policy, which has revealed that stability margin is a function of 1) platoon size (N), 2) ND, 3) IFT, 4) the DC structure [16][30]-[34].

By considering ND as a point mass, Barooah *et.al* (2009) demonstrated that the stability margin approached zero as $O(1/N^2)$ under symmetric bidirectional control, and proved that asymptotic behavior of stability margin could be improved to $O(1/N)$ by introducing small amounts of "mistuning" [16]. This result was extended to linear third-order dynamics, which covers the inertial delay of powertrain dynamics [30]. Using partial differential equation (PDE) approximation, Hao *et al.* (2011) showed that scaling law of stability margin could be improved to $O\left(1/N^{2/D}\right)$ under D-dimensional IFT [32]. Recently, it was proved that employing asymmetric control, the stability margin could be bounded away from zero, which was independent with the platoon size N in [33]. From the perspective of topology selection and control adjustment, Zheng *et al.* (2014) further pointed out two basic methods to improve the stability margin [34].

C. Coherence behavior

The coherence behavior is a scalar metric adopting $\mathcal{H}_2$-norm of the closed-loop system to character the robustness of vehicular platoon driven by exogenous disturbances, which captures the notion of coherence [53][66]. Bamieh *et al.* investigated the asymptotic scaling of upper bounds on coherence behavior with respect to platoon size, and indicated the IFT may play a more important role than DC [66]. Several recent research used coherence behavior as the cost function to optimize the local control gains, *e.g.* in [49], and the communication structure of IFTs, *e.g.* in [50][51][53] by using augmented Lagrangian approach and alternative direction method of multipliers.

V. Discussions

This paper presents a review on vehicular platoon control from the perspective of networked control system. Although some theoretical and/or experimental results have been provided, there are still many open questions, especially considering the emerging large-scaled applications of V2V and V2I communication. Some of them are briefly discussed here:

1) How to model, analyze and design vehicular platoon control in a systematic way. The future challenge comes from the nonlinearity of node dynamics, the variety of topologies, and the low-cost demands of controllers. The communication issues, *e.g.* data delay, quantization error and packet loss, also pose significant challenge on platoon control technique. One interesting question is how to optimize the design of topologies and controllers considering both platoon performances and communication issues.

2) How to balance various performances in a platoon, including practical requirements from highway. The balance of string stability, stability margin and coherence behavior is attracting increasing interests now. Moreover, final goal of platooning is to enhance highway safety, improve traffic utility and reduce fuel consumption. How to consider practical performance requirement is rather challenging for platoon control. A comprehensive design framework is needed to incorporate and balance multiple performance metrics.

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